

Marwan Tawfeek

Data Scientist

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Swedish student residence permit, valid to May 2027

SUMMARY

MSc student in Data Science, Machine Learning and Statistics at Uppsala University on a full funded Swedish institute scholarship, with +3 years of industrial experience in data science, machine learning and growth strategies (credit scoring, customer segmentation, predictive modeling). Currently seeking a thesis opportunity in applied machine learning, NLP/LLMs, and predictive modeling, where I can combine academic knowledge with practical industry experience.

EDUCATION

Master's in Data Science, Machine Learning and Statistics - (Ongoing)

Uppsala University, Uppsala, Sweden

August 2025 – May 2027

- **Swedish Institute Scholarship for Global Professionals (SISGP)** — fully-funded.
- **Relevant coursework:** Statistical Machine Learning, Reinforcement Learning, Deep Learning, Time Series Analysis, Data Engineering I, Database Design I, Scientific Computing for Data Analysis, Artificial Intelligence, LLMs, Mathematical Modeling of Football.

Bachelor's Degree in Computer and Communication Engineering - (A-)

Mansoura University, Mansoura, Egypt

August 2018 - July 2023

PROJECTS

- **GARCH Volatility Modeling of Bitcoin** (Time Series Analysis, solo) — Fitted GARCH(1,1) with Student- t innovations to 13 years of BTC log-returns; validated via Ljung-Box tests and rolling one-step-ahead forecasts.
- **Bike-Share Demand Prediction** (Statistical Machine Learning, group) — Compared 8 classifiers for imbalanced demand-spike prediction (18% positive class); owned Logistic Regression and KNN implementation and tuning; best model (AdaBoost) achieved $F1=0.811$.
- **Distributed Reddit Data Pipeline** (Data Engineering I, group) — Built Spark/HDFS pipeline on a 19.6GB, 3.8M-post dataset across a 4-node OpenStack cluster; ran horizontal/vertical scalability experiments, analyzed results against Amdahl's Law.

WORK EXPERIENCE

Senior Growth Analyst

Prepit, Egypt (Loyalty Programs, F&B) -Startup-

April 2025 – August 2025

- Built a K-means clustering model on customer purchase behavior data. **Enhanced targeting and campaigns conversions by 30%.**
- **Reduced customer churn rate from 65% to 45-50%** through predictive modeling and customized reactivation campaigns.
- Initiated and led the launch of customer badges and gamification mechanics to enhance loyalty and engagement **adopted by 80% of partner restaurants.**

Data Scientist

ADVA, Egypt (Consumer Finance, Fintech) -Startup-

December 2022 – April 2025

- Developed a credit scoring model and routing system using XGBoost, **achieved F1-score 74%.**

- Designed a Quality System Tool with Google Sheets and Python, **reducing data entry errors by 85%**.
- Created a Call Management System in Python for lead prioritization and scoring, **boosting lead conversion rates by 36%**.
- Enhanced ETL processes using Apache Airflow, improving data quality and process efficiency.
- Effectively communicated complex data insights to diverse audiences, including executives.

Data Analyst

*ADVA, Egypt (Consumer Finance, Fintech) -Startup-
December 2021 - December 2022*

- Developed a performance tracking tool using Python and 3CX APIs, **cutting after-call work time by 50%**.
- Automated data scraping using Selenium, improving data acquisition efficiency.
- Developed and maintained performance tracking dashboards using Power BI and Google Sheets.
- Delivered recurring and ad-hoc analyses in Python for cross-functional teams.

SKILLS

- **Programming & Languages:** Python (Numpy, Pandas, Scikit-learn), SQL, Bash
- **Cloud & Deployment:** AWS, Docker, Model Deployment, Git
- **Data Science & ML:** XGBoost, Predictive Modeling, Clustering, Recommendation Systems
- **Statistics & Modeling:** Hypothesis Testing, Time Series (ARCH/GARCH), Cross-Validation, Model Evaluation, Ensemble Methods
- **Data Engineering:** ETL, Data Pipelines, Automation, Web Scraping, Data Validation
- **Visualization & BI:** Power BI, Dashboards, Data Storytelling

VOLUNTEERING/CIVIL SOCIETY ENGAGEMENT

Co-Founder People of Data – Middle East

January 2022 - Present

- Spearheaded the growth of the largest data and AI community in Egypt, connecting 2500+ engineers with business leaders.
- Organized events and workshops to facilitate collaboration and knowledge sharing, leading to valuable partnerships with top tech companies.

Conference of Parties (COP27) - UN Volunteer, Egypt

October 2022 - November 2022

- Assisted the UN Secretariat in quality control of the COP27 digital platform, ensuring high standards for visuals, audio, and live streaming.

LANGUAGES

- Swedish – Beginner (A1) – Actively Learning -
- English – Fluent (C1)
- Arabic – Native